RAJALAKSHMI
ENGINEERING
COLLEGE

DEPARTMENT OF INFORMATION TECHNOLOGY

LAB MANUAL

CS23432 – Software Construction

(REGULATION 2023)

RAJALAKSHMI ENGINEERING COLLEGE

Thandalam, Chennai-602015

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Register No: 2116241001080

Year / Branch / Section: II / IT / B

Semester: IV

Academic Year: 2025-2026

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An AutonomousInstitution

BONAFIDE CERTIFICATE

Name: A.HARISH RAGHAVA

Academic Year: 2025-2026 Semester: IV Branch: IT

Register No 2116241001080

Certified that this is the bonafide record of work lone by the above student in the CS23432 - SOFTWARE CONSTRUCTION *Laboratory*

during the academic year 2025-2026

Signature of Faculty in-charge

Submitted for the Practical Examination held on.................................

Internal Examiner

External Examine

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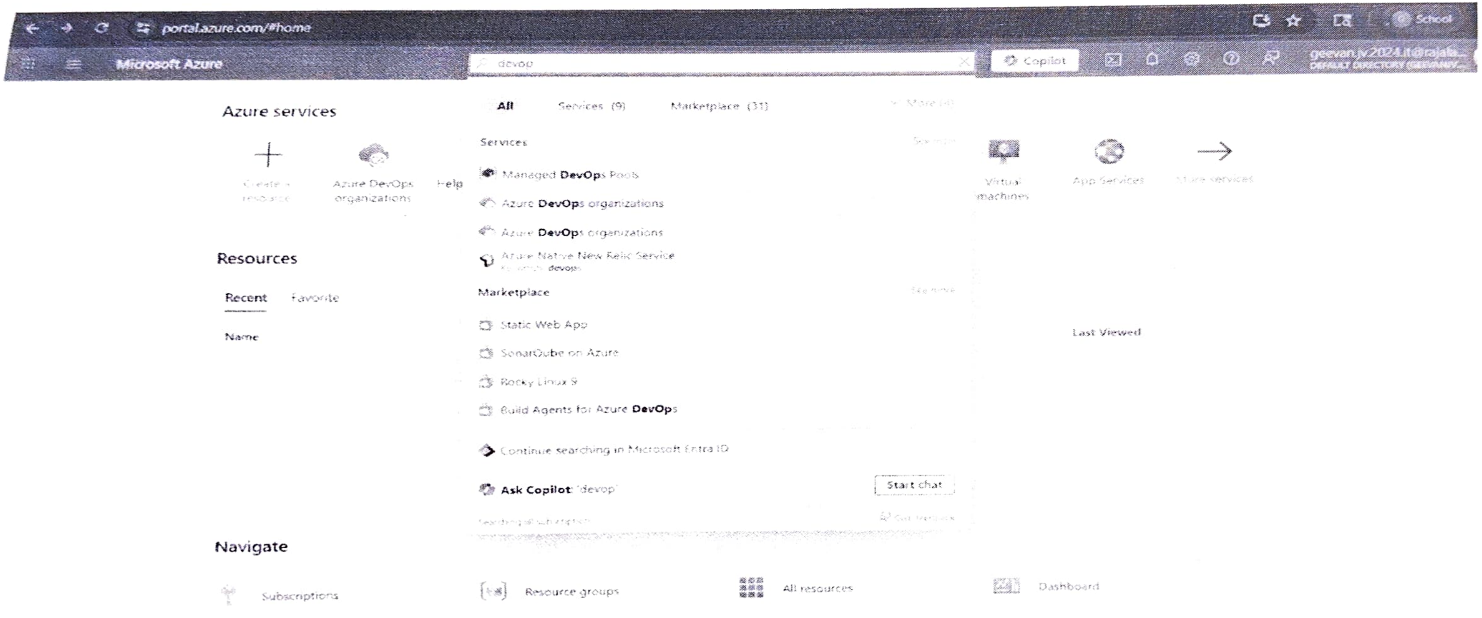

portal.azure.com/#home
Microsoft Azure
devop
Copilot
School
Azure services
All
Services (9)
Marketplace (31)
Services
Managed DevOps Pools
Azure DevOps organizations
Azure DevOps organizations
Azure Native New Relic Service
Marketplace
Static Web App
SonarQube on Azure
Rocky Linux 9
Build Agents for Azure DevOps
Continue searching in Microsoft Entra ID
Ask Copilot 'devop'
Start chat
Azure DevOps organizations
Help
Virtual machines
App Services
Resources
Recent
Favorite
Name
Last Viewed
Navigate
Subscriptions
Resource groups
All resources
Dashboard

EXP NO: 1

AZURE DEVOPS ENVIRONMENT SETUP

AIM:

To set up and access the Azure DevOps environment by creating an organization for the online Quiz project.

PROCEDURE:

1. Open a Web browser and go to the Azure Portal (https://portal.azure.com ↗).
2. Sign in using your Microsoft account credentials. If you do not have an account, create one.
3. In the search bar, type Azure DevOps Organizations and select it.
4. Click on My Azure DevOps Organization.
5. Click Create Organization and provide required details such as:
    * Organization name
    * Region
6. Accept the terms and conditions and create the organization.
7. Once created, navigate to the organization dashboard.
8. Verify the services like Boards, Repos, and Pipelines are accessible.

azure.microsoft.com/en-us/get-started/azure-portal

Ask Google

School

Action required

Microsoft | Azure Explore Products Solutions Pricing Partners Resources

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Microsoft Azure portal

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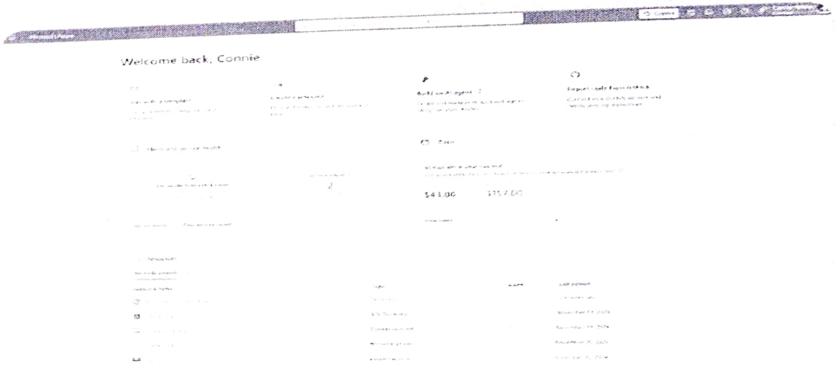


Overview | Azure Copilot | Microsoft Azure mobile app | Next steps

Get started with Azure

RESULT:

Successfully created and accessed Azure DevOps environment.

EXP NO: 2

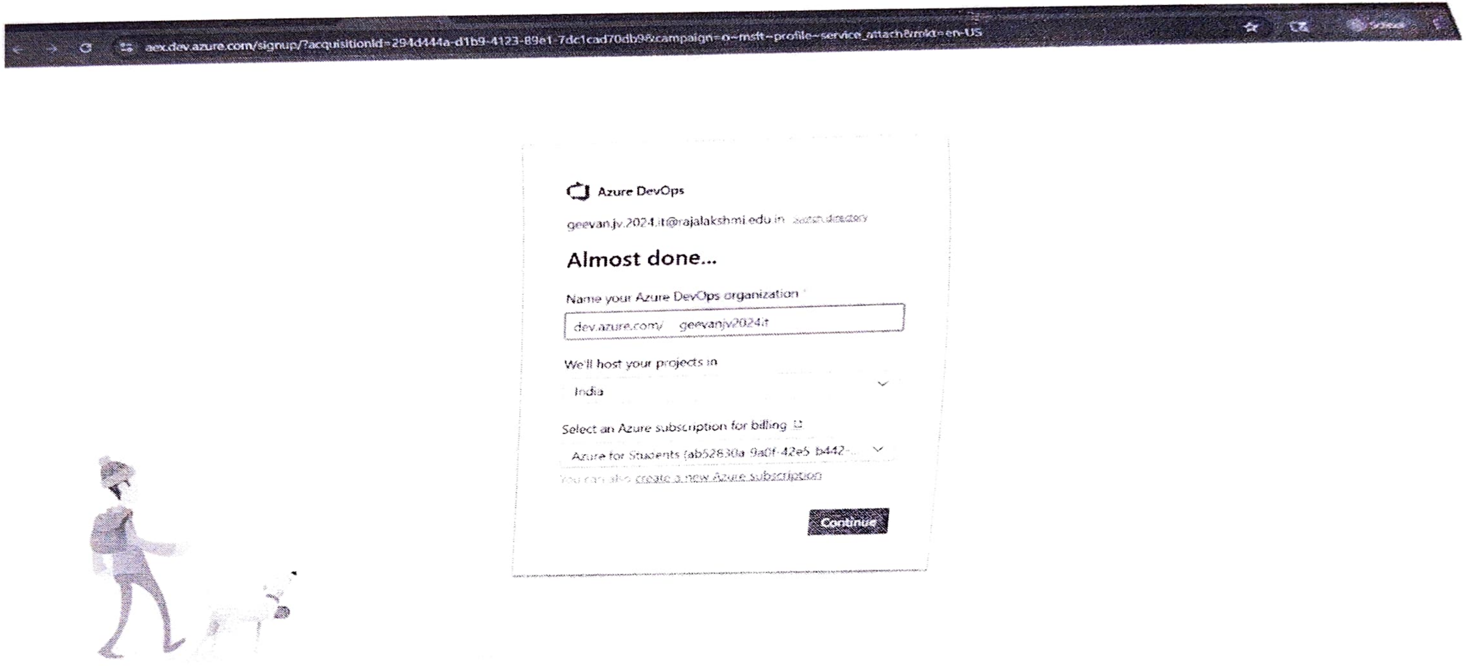
AZURE DEVOPS PROJECT SETUP AND USER STORY MANAGEMENT

AIM :-

To create a project in Azure DevOps and manage user stories for the Online Quiz System.

PROCEDURE :-

1. Login to Azure DevOps and select your Organization.
2. click on New project.
3. Enter project details :
   * Project Name : Online Quiz
   * Description : Web-base Quiz application.
   * Visibility : private
4. Click Create.
5. Navigate to Boards → Work items
6. Click New Work item → User story.
7. Create user stories such as :
   * User registration and login
   * Attempt quiz
   * View results
8. Assign priority, description, and tags to each story.
9. Save and organization the user stories.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Azure DevOps
geevan.jv.2024.it@rajalakshmi.edu.in
Almost done...
Name your Azure DevOps organization
dev.azure.com/ geevanjv2024it
We'll host your projects in
India
Select an Azure subscription for billing
Azure for Students (ab52830a-9a0f-42e5-b442-...
create a new Azure subscription
Continue

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Microsoft harish raghava A Sign out

HA

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harishraghava.a.2024.it@rajalakshmi.edu.in

Microsoft account

India

harishraghava.a.2024.it@rajalakshmi.edu.in

Visual Studio Dev Essentials

Use your benefits

Azure DevOps Organizations

Create new organization

dev.azure.com/harshithkumars2024it

Projects

Online Quizz

Actions

Open in Visual Studio

Manage security

Browse extensions

Leave

RESULT:

Successfully created project and managed user stories.

dev.azure.com/harshithkumars2024/Online%20Quizz/_workitems/recentlyupdated/

Azure DevOps harshithkumars2024 / Online Quizz / Boards / Work items

Online Quizz

- Overview
- Boards
  - Work items
  - Boards
  - Backlogs
  - Sprints
  - Queries
  - Delivery Plans
  - Analytics views
- Repos
- Pipelines
- Test Plans
- Artifacts

Project settings

Work items

Recently updated + New Work Item | Open in Queries | Column Options | Import Work Items | Recycle Bin

Filter by keyword — Types | Assigned to | States | Area | Tags

ID	Title	Assigned To	State	Area Path	Tags
13	Admin Dashboard	Geevan J V	Active	Online Quizz	
11	Dynamic Quiz Generation	Kamal Sundar	Active	Online Quizz	
7	User Registration	Harshith Kumar Suresh	Resolved	Online Quizz	
12	Timer Based Quiz	harish raghava A	New	Online Quizz	
10	View Results	Geevan J V	New	Online Quizz	
9	Attempt Quiz	harish raghava A	New	Online Quizz	
8	Join Quiz	Kamal Sundar	New	Online Quizz	
6	User Registration & Login	Harshith Kumar Suresh	New	Online Quizz	
5	Multiplayer Quiz System	Geevan J V	New	Online Quizz	
4	Quiz Gamification System	Harshith Kumar Suresh	New	Online Quizz	
3	Online quiz	harish raghava A	New	Online Quizz	
	Smart Disaster Response System	Kamal Sundar	New	Online Quizz	

EXP NO : 3

SETTING UP EPICS, FEATURES, AND USER STORIES

AIM :-

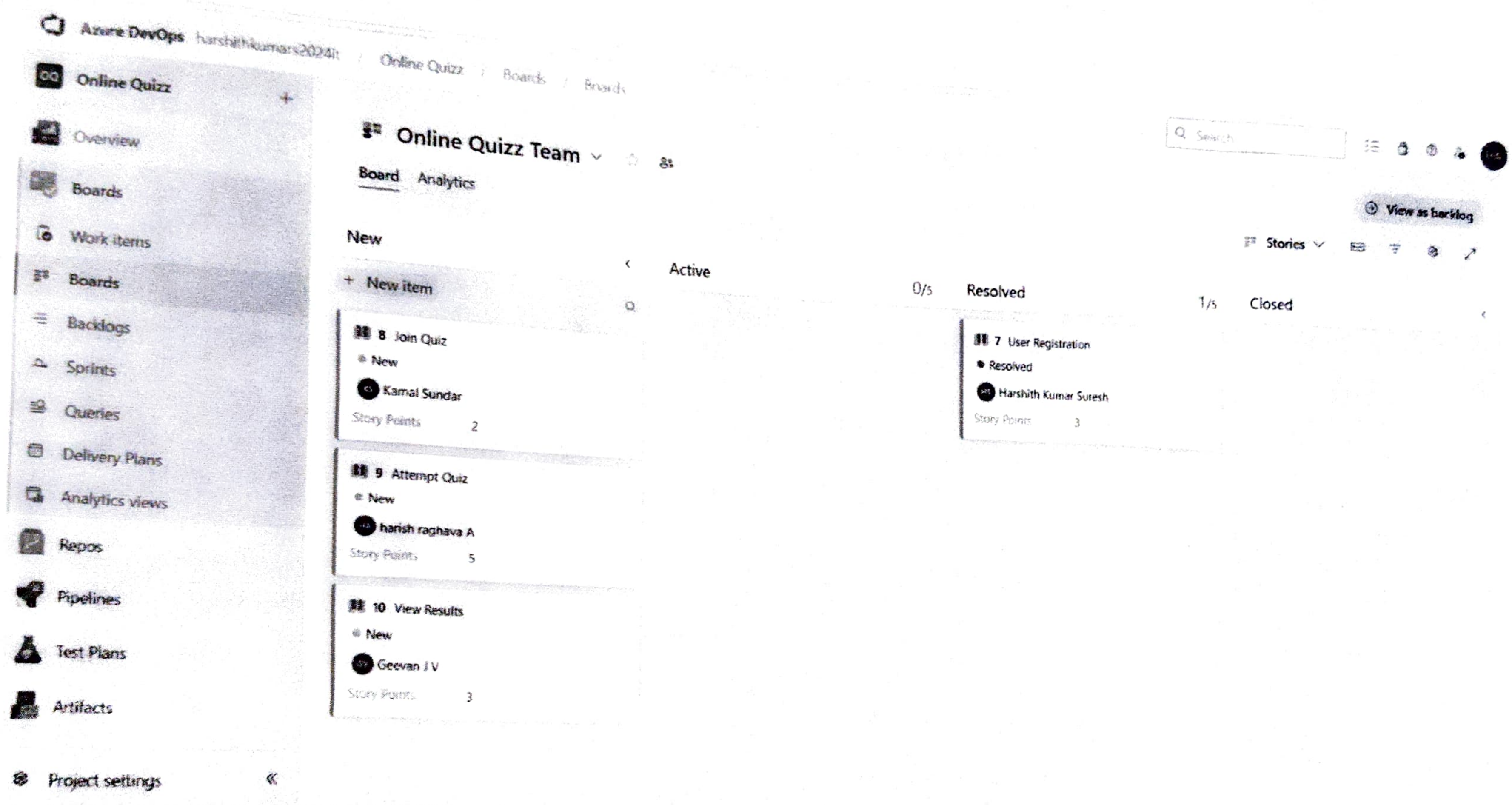
To organize project requirements using epics, features, and user stories.

PROCEDURES :

1. Go to Boards → Backlogs.
2. Create Epics:
   * User Management
   * Quiz Management
3. Under each epic, create Features:
   * Login & Registration
   * Quiz creation
4. Under features, create user stories:
   * User can register
   * User can take quiz
   * User can view results
5. Link user stories to corresponding features and epics.
6. Assign priorities and effort estimates

RESULT:

Successfully structured project using epics, features, and user stories.



Azure DevOps
harshithkumars2024it
Online Quizz
Boards
Boards
Online Quizz
Overview
Boards
Work items
Boards
Backlogs
Sprints
Queries
Delivery Plans
Analytics views
Repos
Pipelines
Test Plans
Artifacts
Project settings
Online Quizz Team
Board
Analytics
View as backlog
Stories
New
New item
Active
0/5
Resolved
1/5
Closed
8 Join Quiz
New
Kamal Sundar
Story Points 2
9 Attempt Quiz
New
harish raghava A
Story Points 5
10 View Results
New
Geevan J V
Story Points 3
7 User Registration
Resolved
Harshith Kumar Suresh
Story Points 3

EXP NO: 4

SPRINT PLANNING

AIM :-

To plan and assign work into sprints for the online Quiz project.

PROCEDURE :-

1. Navigate to Boards → Sprints
2. Create multiple sprints:
   * Sprint 1: Login & Registration
   * Sprint 2: Quiz creation
   * Sprint 3: Quiz Attempt
   * Sprint 4: Result processing
3. Define sprint duration (e.g., 2 weeks).
4. Drag and drop user stories into respective sprints.
5. Assign tasks to team members.
6. Set sprint goals and timelines.
7. Monitor progress using sprint board.

RESULT:

Sprints successfully planned and user stories assigned.

Online Quizz   Boards   Work items

Recently updated

Back to Work Items

USER STORY 9

9 Attempt Quiz

harish raghava A

0 Comments   Add Tag

New

Area: Online Quizz

Iteration: Online Quizz\Iteration 1

7 of 12

Follow

Details

Description

As a user, I want to answer quiz questions within a time limit so that I can complete the quiz

Acceptance Criteria

Discussion

Planning

Story Points
5

Priority
3

Risk
3 - Low

Classification

Value area
Business

Deployment

To track releases associated with this work item, go to Releases and turn on deployment status reporting for Boards in your pipeline's Options menu. Learn more about deployment status reporting

Development

Add link

Link an Azure Repos commit, pull request or branch to see the status of your development. You can also create a branch to get started.

XP NO: 5

POKER ESTIMATION

AIM:

To estimate the effort required for user stories using poker estimation technique.

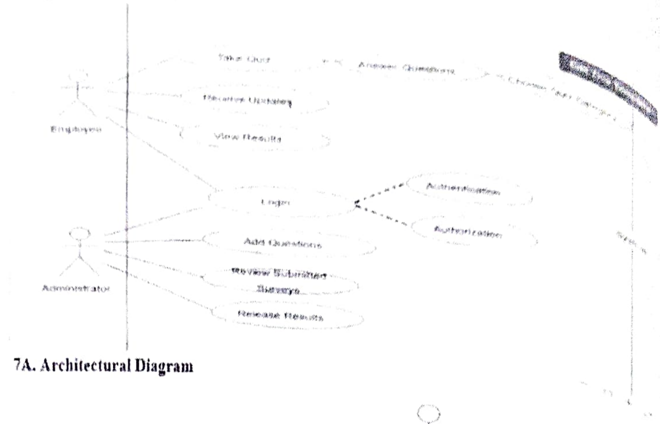
PROCEDURE:

1. Select user stories from backlog.
2. Conduct estimation session with team members.
3. Use fibonacci scale (1, 2, 3, 5, 8, 13).
4. Assign story points:
   * Login → 3
   * Quiz Attempt → 5
   * Result Generation → 8
5. Discuss differences in estimation among team members.
6. Finalize story points after consensus
7. Update estimates in Azure Devops.

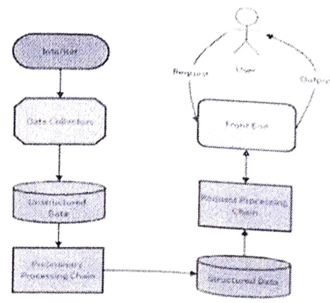
Result:

Successfully estimated effort for user stories.

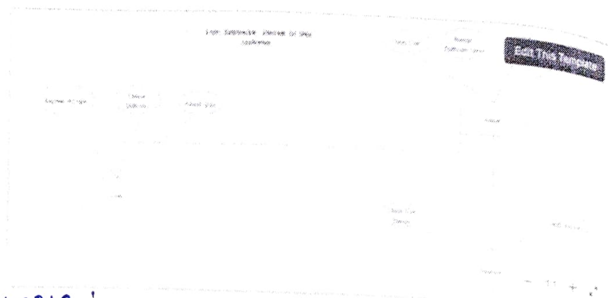
Class Diagram :



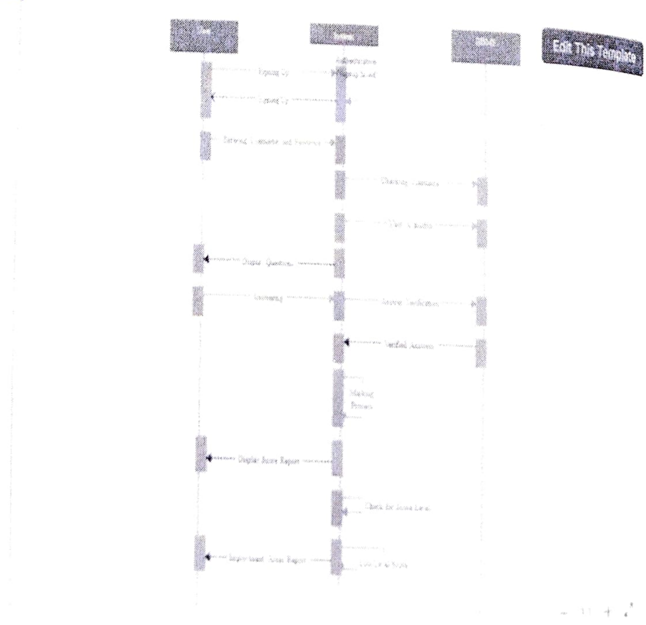
7A. Architectural Diagram



7B.ER Diagram



Sequence :



EX NO : 6

CLASS AND SEQUENCE DIAGRAMS

AIM:

To design class and sequence diagrams for the Online Quiz systems.

Procedure:

Identify main classes:

* User
* Quiz
* Question
* Result

2. Define attributes and methods for each class.
3. Establish relationships b/w classes.
4. Draw the class diagram.
5. For sequence diagram:
   * Identify interaction flow
   * Define actors (user, system)
6. Draw sequence steps:
   * Login → Select Quiz → Answer Questions → Submit → View Result.
7. Use diagram tools like Draw.io of UML tools.

Created

EX.NO: 7

Date:

ARCHITECTURAL AND ER DIAGRAMS

AIM:

TO design system architecture and ER Diagram for online quiz.

Procedure:

1. Identify system components.

* frontend (UI) * Backend (server) * Database

2. Draw Architectural diagram showing interaction b/w components

3. Identify database entities:

* User * Quiz * Question * Result

4. Define attributes for each entity

5. Establish relationships

* User attempts Quiz
* Quiz contains Question

6. Draw ER Diagram using tools

RESULT:

Architectural and ER diagrams designed successfully.

EX NO: 8

Date

TEST PLANS AND TEST CASES

AIM:

TO design test plans and test cases for the online quiz system.

PROCEDURE:

1. Identify core features:
   * login * Quiz attempt * Result Display
2. Create test plan in Azure DevOps
3. Create test suites based on Modules
4. Design test cases:
   * Happy path (valid inputs)
   * Error path (invalid inputs)
5. Example:
   * TC01 : Valid login → Success
   * TC02 : Invalid login → Error
6. Define steps and Expected results
7. Execute test cases and record results
8. Report bugs if any.

RESULT:

Test plans and test cases created Successfully.

EXP NO: 9 LOAD TESTING AND PIPELINES

AIM:-

TO perform load Testing and create CI/CD Pipeline for online quiz.

Procedure :-

Load Testing :-

1) GO to Azure portal
2) create Azure Load Testing resource
3) provide application URL
4) Configure number of users
5) Run test and analyze performance

Pipeline :-

1) Connect Github repository to Azure Devops
2) Create pipe line using YAML file
3) Add steps :-
   * Install dependies
   * Run application
4) commit code to bigger pipeline
5) Monitor pipeline execution

Result :

Load testing and pipeline executed successfully.

Exp No: 10 GITHUB PROJECT STRUCTURE AND NAMING CONVENTIONS

AIM :-

TO define project structure and naming Conventions for online quiz systems.

Procedure:

1) create Github repository
2) organize folders.

Code:

```
Online quiz/
 ├ frontend/
 ├ backend/
 ├ database/
 ├ tests/
 └ README.md
```

3) Follow naming conventions :-

* files → lower case
* classes → pascal case
* variables → camel case

4) Add README file with project details
5) Maintain proper documentation.

RESULT :

project structure and naming conventions defined Successfully